

Introduction

Most of us are familiar with the concepts of reliability and MTBF at a superficial level, without considering what lies behind the figures quoted and what significance should be attached to them. The subject deserves a deeper understanding, so, let us start by having a better look at the terminology.

1.1 Failure Rate (λ)

The failure rate is defined as the percentage of units failing per unit time. This varies throughout the life of the equipment and if (λ) is plotted against time, the characteristic "bathtub" curve is obtained for most electronic equipment (See Figure 1).

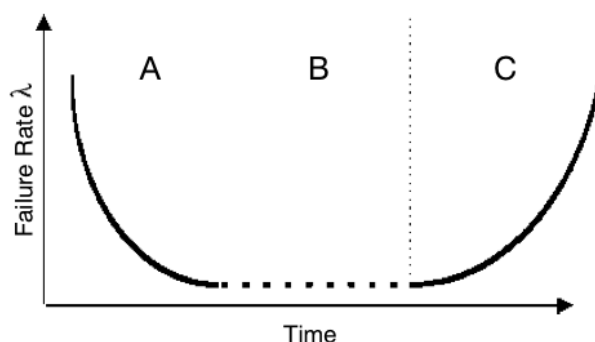


Fig 1. Failure Rate vs. Time

This curve has three regions:

- A Infant mortality.
- B Useful life.
- C Wear out

In region "A", poor workmanship and substandard components cause failures. This period is usually a few hundred hours and a "burn in" is sometimes employed to stop these failures occurring in the field. Note that this does not stop the failures occurring, it just ensures that they happen in-house and not on the customer's premises.

In region "B", is approximately constant and it is only for this region that the following analysis applies.

In region "C", components begin to fail through having reached their end of life, rather than by random failures.

Examples are electrolytic capacitors drying out, fan bearings seizing up, switch mechanisms wearing out etc. Well implemented preventive maintenance can delay the onset of this region.

1.2 Reliability ($R(t)$)

There are a large number of definitions and one will get different answers from statisticians, engineers, mathematicians and so on, an essentially practical definition is: The probability that a piece of equipment operating under specified conditions shall perform satisfactorily for a given period of time.

Probability is involved since it is impossible to predict the behaviour with absolute certainty. The criterion for "satisfactory performance" must be defined as well as the operating conditions such as input, output, temperature, load etc.

1.3 Mean Time Between Failures (MTBF), Mean Time To Failure (MTTF)

Strictly speaking, MTBF applies to equipment that is going to be repaired and returned to service, MTTF to parts that will be thrown away on failing.

The MTBF is the inverse of the failure rate.

$$MTBF = \frac{1}{\lambda} \quad \dots(1)$$

Many people, unfortunately, misunderstand MTBF, and tend to assume that the MTBF figure indicates a minimum, guaranteed, time between failures. This assumption is wrong, and for this reason the use of the failure rate rather than the MTBF is highly recommended.

$$R(t) = e^{-\lambda t} = e^{-\frac{t}{m}} \quad \dots(2)$$

$$m = \frac{t}{\log_n \left(\frac{1}{R(t)} \right)} \quad \dots(3)$$

Where

$R(t)$ = Reliability
 e = Exponential (2.718)
 λ = Failure Rate
 m = MTBF

Note that for a constant failure rate, plotting reliability against time "t" gives a negative exponential curve (See Figure 2).

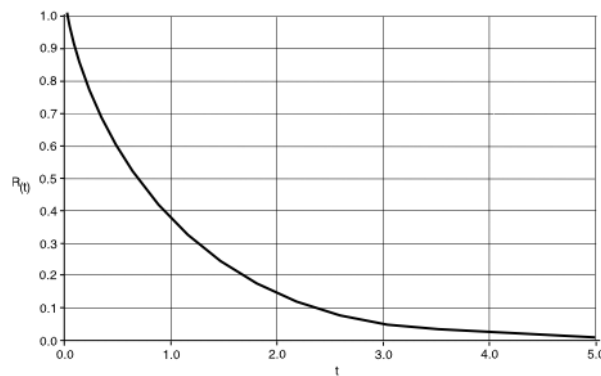


Fig 2. Reliability $R(t)$ Plotted Against (λt) for a unit with a constant failure rate λ

When $t/m = 1$, i.e., after a time "t", numerically equal to the MTBF figure "m":

$$R(t) = e^{-1} = 0.37 \quad \dots(4)$$

Equation (4) can be interpreted in a number of different ways:

- (a) If a large number of units are considered, only 37% of them will survive for as long as the MTBF figure.
- (b) For a single unit, the probability that it will work for as long as its MTBF figure, is only 37%.
- (c) We can say that the unit will work for as long as its MTBF figure with a 37% Confidence Level.

In order to put these numbers into context, let us consider a power supply with a MTBF of 500,000 hours, (a failure rate of 0.2%/1000 hours), or as the advertising would put it "an MTBF of 57 years!"

From eq.(2), $R(t)$ for 26,280 hours (3 years) is approximately 0.95, i.e., if such a unit is used 24 hours a day for 3 years, the probability of it surviving that time is 95%. The same calculation for a ten year period will give a $R(t)$ of 84%.

Now let us consider a customer who has 700 such units. Since we can expect, on average, 0.2% of units to fail per 1000 hours, approximately one unit per month will fail on average, since the number of failures per year is:

$$\frac{0.2}{100} * \frac{1}{1000} * 700 * 24 * 365 = 12.26$$

1.4 Service Life (Mission Life, Life)

Note that there is no direct connection or correlation between service life and failure rate. It is perfectly possible to design a very reliable product with a short life. A typical example is a missile for example: it has to be very, very reliable (MTBF of several million hours), but its service life is only 0.06 hours (4 minutes). 25 year old humans have a MTBF of about 800 years, (FR about 0.1%/year), but not many have a comparable "service life". Just because something has a good MTBF, it does not necessarily have a long service life as well (See Figure 3).

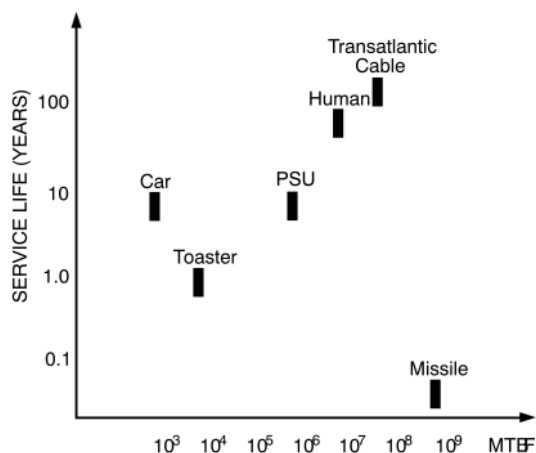


Fig 3. Examples of Service Life vs. MTBF

Factors Affecting Reliability

2.1 Design Factors

The most important factor is good, careful design based on sound experience, resulting in known safety margins. Unfortunately, this does not show up in any predictions, since they assume a perfect design!

It has to be said that a lot of field failures are not due to the classical random failure pattern discussed here, but to shortcomings in the design and in the application of the components, as well as external factors such as occasional voltage surges, etc. These may well be 'outside specification' but no one will ever know, all that will be seen is a failed unit. Making the units rugged through careful design and controlled overstress testing is a very important part of making the product reliable.

The failure rate of the equipment depends on three other factors:

- Complexity
- Stress
- Inherent (generic) reliability of the components used

2.2 Complexity

Keep things simple - what isn't there, can't fail, but be careful: what isn't there can cause a failure! A complicated or difficult specification will, invariably result in reduced reliability. This is not due to the shortcomings of the design staff, but to the resultant component count. Every component used will contribute to the equipment's unreliability.

2.3 Stress

In electronic equipment, the most prominent stresses are temperature, voltage, vibration, and temperature rise due to current. The effect of each of these stresses on each of the components must be considered. In order to achieve good reliability, various derating factors have to be applied to these stress levels. The derating has to be traded off against cost and size implications.

Great care and attention to detail is necessary to reduce thermal stresses as far as possible. The layout has to be such that heat-generating components are kept away from other components and are adequately cooled.

Thermal barriers are used where necessary and adequate ventilation needs to be provided. The importance of these provisions cannot be overstressed since the failure rate of some components will double for a 10 °C increase in temperature. Note that decreasing the size of a unit without increasing its efficiency will make it hotter, and therefore less reliable!

2.4 Generic (Inherent) Reliability

Inherent reliability refers to the fact that film capacitors are more reliable than electrolytic capacitors, wirewrap connections more reliable than soldered ones, fixed resistors more reliable than pots, and so on. Components have to be carefully selected to avoid the types with high generic failure rates. Quite often, there is a cost trade off - more reliable components are usually more expensive.

3. Estimating the Failure Rate

The Failure Rate should be estimated and measured throughout the life of the equipment:

- During design, it is predicted
- During manufacture, it is assessed
- During the service life, it is observed

3.1 Prediction

Predicting the failure rate is done by evaluating each of the factors effecting reliability for each component and then summing these to get the failure rate of the whole equipment. It is essential that the database used is defined and used consistently. There are three databases in common use: MIL-HDBK-217, HRD5 and Bellcore. These reflect the experiences of the US Navy, British Telecom and Bell Telephone, respectively.

Other sources of data are component manufacturers and some large companies like Siemens, Philips, France Telecom or Italtel. Data from these should not be used unless specifically requested by the customer.

In general, predictions assume that:

- The design is perfect, the stresses known, everything is within ratings at all times, so that only random failures occur
- Every failure of every part will cause the equipment to fail.
- The database is valid

These assumptions are wrong. The design is less than perfect, not every failure of every part will cause the

equipment to fail, and the database is likely to be at least 15 years out-of-date. However, none of this matters much, if the predictions are used to compare different topologies or approaches rather than to establish an absolute figure for reliability. This is what predictions should be used for.

3.1.1 Parts Stress Method

In this method, each factor affecting reliability for each component is evaluated. Since the average power supply has over 100 components and each component about 7 factors (Typically: stress ratio, generic, temperature, quality, environment, construction, and complexity) this method requires a considerable effort and time. Predictions are usually done in order to compare different approaches or topologies, i.e. when detailed design information is not available and the design itself is still in a fluid state. Under such circumstances, it is hardly worthwhile to spend this effort, and the much simpler and quicker Parts Count Method is used.

3.1.2 Parts Count Method

In this method, all like components are grouped together, and average factors allocated for the group. So, for example instead of working out all the factors for each of the 15 electrolytic capacitors used, there is only one entry of cap. electr.' and a quantity of 15. Usually only two factors are allocated: generic and quality. The other factors, including stress levels, are assumed to be at some realistic level and allowed for in the calculation. For this reason, the factors are not interchangeable between the two methods. In general, for power supplies, HRD5 gives the most favourable result, closely followed by Bellcore, with MIL-HDBK-217F the least favourable. This depends on

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the mix of components in the particular equipment, since one database maybe "unfair" on ICs, and an other on FETs. Hence, the importance of comparing results from like databases only.

3.2 Assessment

This is the most useful and accurate way of predicting the Failure Rate. A number of units are put on "life test" (more correctly described as a Reliability Demonstration Test), usually at an elevated temperature, and so the stresses and the environment is controlled. Note, however, that it is not always possible to model the real environment accurately in the laboratory.

During life-tests and reliability demonstration tests, it is usual to apply greater stresses than normal, so that we get to the desired result quicker. Great care has to be applied to ensure that the effects of the extra stress is known and proven to be calculable, and that no hidden, additional failure mechanisms are activated by the extra stress. The usual "extra stress" is an increase of temperature, and its effect can be calculated from the Arrhenius equation, as long as the maximum ratings of the device are not exceeded.

Note that the accelerating effect depends on the activation energy that applies for the chemistry of the particular component. This would indicate that the Acceleration Factor from 25 °C to 50 °C is approx. 5.25, so be suspicious of results based on 0.7 eV, or even 1 eVn.

At the beginning of such a test it is sometimes difficult to distinguish between early failures ("infant mortality", region A) and the first failures belonging to the "constant failure rate" region (region B). In such cases, the Cumulative Distribution Function is plotted on Weibull paper. This paper has double logarithmic scaling such that a constant failure rate will result in a straight line at an indicated gradient of 1. Decreasing FR (region A) will give a smaller gradient, increasing FR (wearout, region C) a higher gradient. Both the available time and the number of units on test are limited, and so it is of the utmost importance that the maximum amount of useful information is extracted from a limited amount of data. Statistical methods are used to achieve this.

3.2.1 Confidence limits

What we are attempting to do is to predict the behaviour of the large number of units in the field (called the population) from the behaviour of a small number of randomly selected units (called the sample). This process is called Statistical Inference. The results obtained by such means cannot, of course, be completely accurate, and it is therefore essential to establish the degree of accuracy that applies. This is done by estimating the mean value and defining a band or an interval around this estimated mean that will include the actual, true mean value of the complete population. Such an interval is defined by a Confidence Limit, i.e., if we establish that the failure rate is between 1%/1000 hours and 2%/1000 hours with a Confidence Limit of 90%, this means that we expect 90% of the units in the field to exhibit failure rates between these limits, and the other 10% of units to have a lower or higher failure rate. For a population exhibiting a constant failure rate,

$$\lambda = \frac{X^2 (2r+2), (1-\phi)}{2tN} \quad \dots(5)$$

where λ = demonstrated failure rate with a one-sided higher confidence limit of ϕ (phi)

t = test time
N = number of units on test
r = number of failures
 $X^2 (2r+2), (1-\phi)$ = value of the X^2 distribution

λ

r	$\phi = 0.6$	$\phi = 0.9$
0	93×10^3	230×10^3
1	200×10^3	390×10^3
2	310×10^3	530×10^3
3	420×10^3	670×10^3
4	530×10^3	790×10^3
5	630×10^3	910×10^3
6	730×10^3	1040×10^3
7	830×10^3	1160×10^3
8	930×10^3	1300×10^3
9	1040×10^3	1410×10^3
10	1140×10^3	1530×10^3

with probability $(1 - \alpha)$ of not being exceeded in random sampling where $(2r + 2)$ is the number of degrees of freedom.

The constants given by this equation are tabulated below for values of r between 0 and 10, and for α of 0.6 and 0.9. (These are the usual values for Confidence Limits used in industry).

To use this table divide the factor given by the total number of unit-hours to get the failure rate in %/1000 hours. Let us consider the case when we have 50 units on test and one fails after 4 months (2920 hours):

$$t = 2920, N = 50, r = 1$$

From the table, we can say with 60% confidence that the failure rate will be less than:

$$\frac{200,000}{50 \times 2920} = 1.37\%/1000 \text{ hrs}$$

Alternatively, we can say with 90% confidence that the failure rate will be less than:

$$\frac{390,000}{50 \times 2920} = 2.67\%/1000 \text{ hrs}$$

In the parent population, therefore, we expect 60% of the units to exhibit a failure rate better than 1.37%/1000 hrs (an MTBF of 73,000 hrs.), and therefore 40% of units to have a FR worse than that; or 90% of the units to be better than 2.67%/1000hrs (an MTBF of 37,400 hrs.), and therefore 10% of units to have a FR worse than that (See Figure 4).

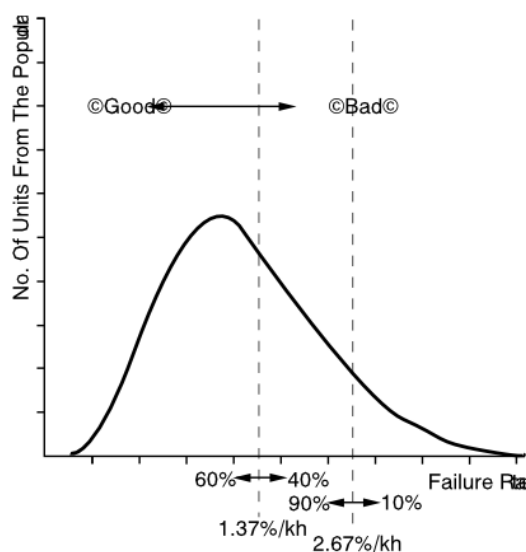


Fig 4. Confidence Limit

However, there is a practical problem with this method: although we get valid answers, the length of time for that answer is a function of

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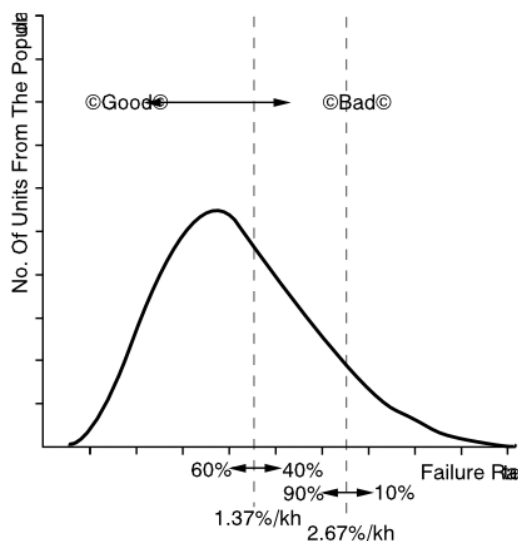


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However, there is a practical problem with this method: although we get valid answers, the length of time for that answer is a function of

1. The sheer mechanics of actually collecting and collating the data.
2. The uncertainty of the duty, conditions of use and stresses, or abuse, that the units were subjected to.

Great care has to be exercised in drawing conclusions due to the difficulty of distinguishing between true random failures and misuse in the field (accidental or otherwise).

4. Prototype Testing

With all the sophisticated computer analysis, simulation and tolerancing methods available, there is still no substitute for thoroughly testing the maximum number of prototypes. An effort should be made to locate and use components from different batches, especially for critical components. These units must be tested under dynamic conditions to ensure reliability. An effective test is to cycle the temperature, the input, and the load independently. The units should be tested at both maximum and minimum temperature cycling according to this plan. Cpk analysis of the results is used to ensure that the specification parameter margins are adequate. After testing, these units are normally used as the first batch on the reliability demonstration tests.

At least one unit should be subjected to HALT testing, and several to destructive overstress tests to establish the safety margins.

The timing of these tests is critical - it must not be so early in the development phase that the final circuit is radically different, and it must not be so late that production starts before the results are evaluated. A pitfall to watch out for, if changes are proposed as a result of these tests is that the up-dated units must be subject to long term testing themselves.

5. Manufacturing Methods

This is a separate subject in itself, but there are three main factors contributing to unreliability in manufacture:

- Suppliers
- Manual assembly methods
- Tweaking of settings and parameters

Suppliers must be strictly controlled to deliver consistently good devices, with prior warning of any process changes and any other changes. These days, with modern QA practices and JIT manufacturing methods, this level of reliability is achieved by dealing with a small number of trusted suppliers. Manual assembly is prone to errors and to some random, unintentional abuse of the components by operators. This creates latent defects, which show up later.

Tweaking produces inconsistency and side effects. A good motto is: if it works, leave it alone; if it does not, find the root cause and do not tweak. There must be a root cause for the deviation, and this must be found and eliminated, rather than masked by the tweak. There are well-established TQM and SPC methods to achieve this. Testing and Quality Assurance has a major part to play. Testing must be appropriate to ensure that the units perform well in the application. Cpk analysis ensures that the specification parameter margins are adequate and controlled.

6. System Reliability

There are two further methods of increasing system reliability. Firstly, more reliable components. MIL standard or other components of assessed quality could be used, but in industrial and commercial equipment, the expense is not normally justified.

Secondly, redundancy. In a system where one unit can support the load, and two units are used in parallel, the system is much more reliable since the system will still work even with one unit failed. Clearly, the probability of two units failing simultaneously is much less than that of one unit failing. This system would have a big size and cost penalty, (twice as big and twice as much) so normally an N+1 system is used, where N units can support the load, but N+1 units are used in parallel, "2+1" or "3+1" being the usual combinations. Supposing the reliability of each unit under the particular conditions is 0.9826, (m=500,000h, t=1year) the system reliability for an "N+1" system where N = 2 would be 0.9991, an improvement of 20 times. (Nearly 60 times in a 1+1 system).

However, there are many pitfalls in the system design, such as:

1. N units must be rated to support full load.
2. Any part failing must not make the system fail.
3. If any part fails this must be brought to the operator's notice so that it can be replaced.
4. Changing units must not make the system fail (hot plugging).

It is very difficult and tricky to design the system to satisfy items 2 & 3. For example the failure of components that do not effect system

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Productivity and process yield

Contents

1	Introduction	1
2	Fab yield	2
3	Wafer sort yield	5

1 Introduction

Process and device evaluation in the fab is used to make sure that the fabrication goals are met, at the individual process level. These goals are usually defined as a *process window* for each step in the fabrication process. This could be a maximum level for defect density and for a growth process a range of acceptable thickness values. Semiconductor IC manufacture is a complex process, with hundreds of individual steps. Errors in even one step have the potential to dramatically affect the functioning of the final product. Even one *killer defect* can cause device failure. The goal of the manufacturing process is to improve the **yield** of the process. There are three major yield measuring parameters

1. **Wafer fabrication yield** or **fab yield** - this is defined as the ratio of the total number of wafers that come out of the fab (after the end of all the individual processes, including measurement) to the total number of wafers that were started in the fab.

Table 1: The major yield measurement points in IC fabrication. There are three yield parameters, wafer yield is defined in the fab while sort and packaging yield are defined for processes after the fab.

Major yield measuring points	
Manufacturing stage	Measured
Wafer fabrication yield	$\frac{\text{Wafers out of fab}}{\text{Wafers started in the fab}}$
Wafer sort yield	$\frac{\text{Functioning dies on wafer}}{\text{Total dies on wafer}}$
Packaging yield	$\frac{\text{Packages passing final testing}}{\text{Good dies started for packaging}}$

2. **Wafer sort yield** - this is the ratio of the number of good (functioning) dies (after electrical testing) to the total number of dies in the wafer.
3. **Packaging yield** - this is the ratio of the total number of packaged dies that pass final electrical testing (e-test) to the number of good dies that are started for packaging. This measures the yield of the packaging process, which is a series of steps similar to fabrication process.

The major yield points are summarized in table 1. The different ways the yield points are related is shown in figure 1.

2 Fab yield

Fab yield or wafer fabrication yield measures the output of the wafer fabrication process. It is a *cumulative yield*, since wafers go through a number of individual steps within the fab. For each step or process, it is possible to define a yield parameter. This is called a **station yield** and is defined as

$$\text{Station yield} = \frac{\text{number of wafers leaving the station}}{\text{number of wafers entering the station}} \quad (1)$$

Thus, the fab yield is defined as the product of the individual station yields, since the output of each step is the input for the next

$$\text{fab yield} = \text{station yield 1} \times \text{station yield 2} \times \text{station yield 3} \times \dots \quad (2)$$

For an overall high fab yield, the individual station yields must also be high. Also, as the number of fab processes increase, the fab yield must decrease

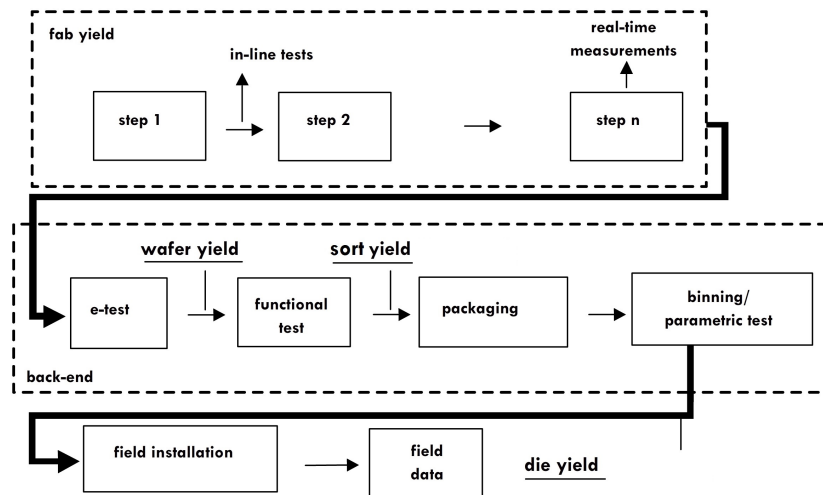


Figure 1: The three yield points and their relation to IC process flow. Fab yield is related to the individual station yields in the fab. Sort yield is related to e-test and die yield is after packaging. Adapted from *Semiconductor manufacturing and process control* - May and Spanos.

since the station yield can never be larger than 1. Consider a sample ten step process as shown in table 2. These are various steps in the formation of a MOSFET. The individual station yields are all shown in percentages. Even though these yields are all high (each station yield is above 90%), the cumulative yield of the process is only 83.9. This decrease in overall yield will become more pronounced as the number of process steps increases. There are various factors that affect the overall fab yield

1. Number of process steps - for a 50 step process to get 75% yield, the individual steps must each have a yield greater than 99.4%.
2. Wafer breakage and warping - this is related to the handling of the wafers. The handling process is now automated in most commercial fabs. Table 3 lists the number of handling steps in an oxidation process. Each step has the potential to lower yield, by causing wafer breakage. Warping happens when wafers are heated to and cooled from high temperatures e.g. rapid thermal annealing process.
3. Process variation - this is related to the process window for each step in the fabrication. This is also called **spec limit**. If the spec limits are too loose, then the process variation can cause lower yield. On the

Table 2: Ten step process with individual station and cumulative yields, listed in percentages. As we go down the process flow, the cumulative yield decreases since the output of each step is the input for the next

Step	Wafers in	Station yield	Wafers out	Cumulative yield
Field ox	1000	99.5	995	99.5
S/D mask-ing	995	97	965	96.5
S/D doping	965	99.3	958	95.8
Gate mask	958	99.0	948	94.8
Gate ox	948	99.0	938	93.
Contact hole mask	938	92.0	862	86.2
Metal layer dep	862	99.0	853	85.3
Metal layer mask	853	100	853	85.3
Alloy metal layer	853	99.5	848	84.8
Passivate	848	99	839	83.9

Table 3: Wafer handling steps in a typical oxidation furnace. Each handling step can cause defects in the wafer affecting yield. The goal is to minimize wafer handling. Adapted from *Microchip fabrication* - Peter van Zant.

Substep	No. of wafer handlings
Wafers are removed from carrier and placed in cleaning boat	2
Wafers are cleaned, rinsed, and dried	1
Wafers are removed from cleaning boat, inspected, and placed on oxidation boat	2
Boat is removed from furnace	0
Wafers are removed from boat and placed back in the carrier	1
Test wafers are removed from carrier and measured	2
Total no. of handlings	8

other hand, too tight a spec limit can lower individual station yield and hence fab yield.

4. Process defects - these are isolated defects that happen during processing. They could be due to equipment issues or incoming wafer issues.
5. Mask defects - these are caused due to defects in the hard mask like dust, cracks in the quartz, damaged Cr layer. Some mask defects are shown in figure 2.

Batch processes should have higher process control (i.e. higher station yield) than serial processes, since more than one wafer is handled at a time. Typically, Si wafers have a higher yield than GaAs, since GaAs is more brittle. For GaAs, even partial wafers are processed in the fab due to their brittle nature, while for Si, partial or broken wafers are discarded.

3 Wafer sort yield

Wafer sort yield refers to the number of functioning (good) dies in the wafer that comes out of the fab to the total number of dies in the wafer. There are a number of factors on which sort yield depends on

1. Wafer diameter - as the wafer diameter increases, number of edge dies decreases. This increases sort yield.
2. Die area - as the individual die area increases, yield decreases. So, larger wafers are more cost effective for larger dies.
3. Number of processing steps - this is related to fab yield in that as the number of process steps increases, the background defect density also increases. The effect of wafer size and die size on the number of wafers is shown in figure 3.
4. Circuit density - this defines the smallest defect size that can cause a wafer to short. Not all defects in a wafer are fatal, but as the circuit density increases, the number of fatal defects increases.
5. Defect density - a high background defect density in the fab can cause dies to fail. This effect is more pronounced for larger dies, as shown in figure 4.
6. Crystal defect density - this relates to the quality of the Si wafer that is used for IC manufacturing. Existing defects like dislocations can be increased during processing, which can affect yield.

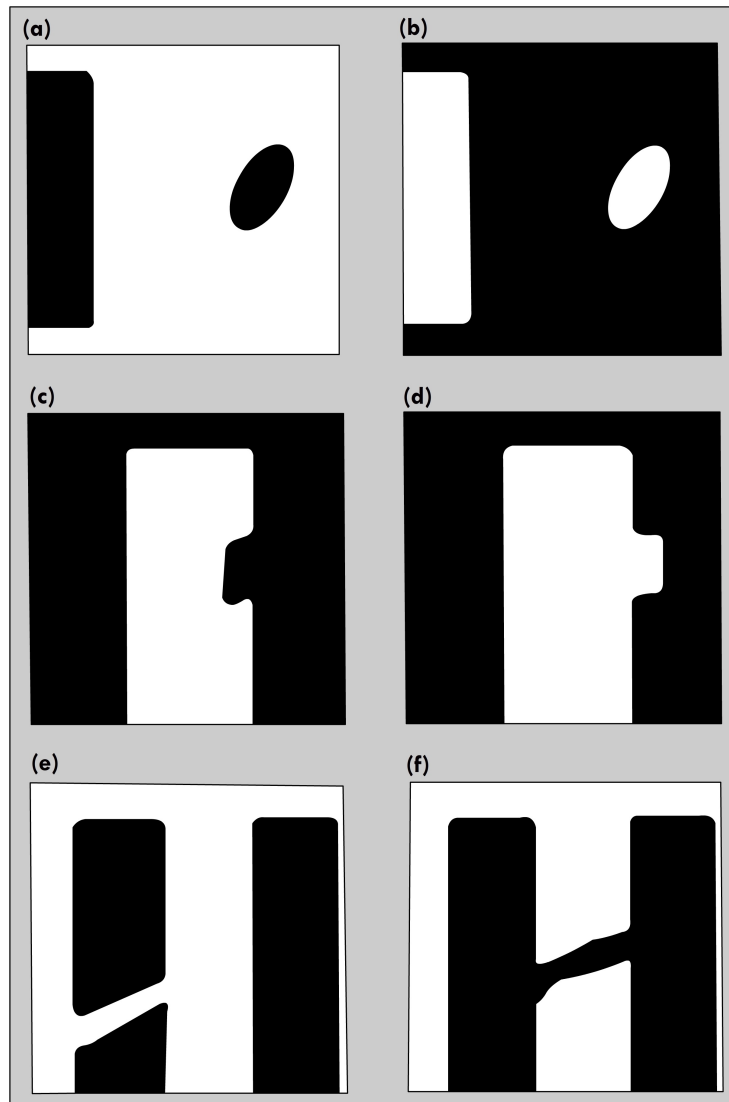


Figure 2: (a) - (f) Typical mask defects that can get transferred to the wafer during lithography. Not all of these defects are killer defects. Adapted from *Microchip fabrication* - Peter van Zant.

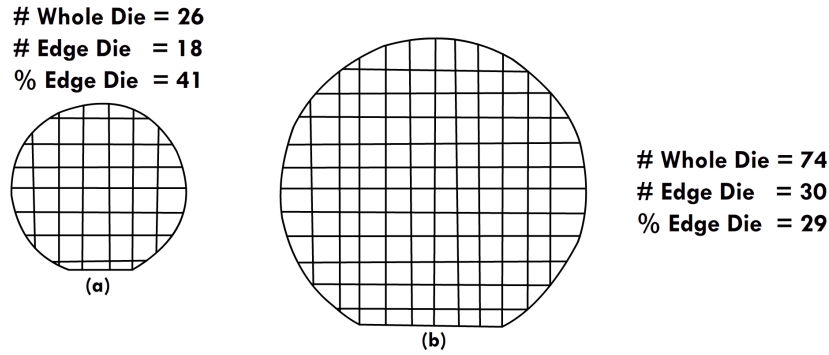


Figure 3: Relation between wafer size and die size effects. (a) Smaller dies have larger number of edge dies (b) Larger wafers have smaller edge dies, for the same die size. This leads to greater yield. Adapted from *Microchip fabrication* - Peter van Zant.

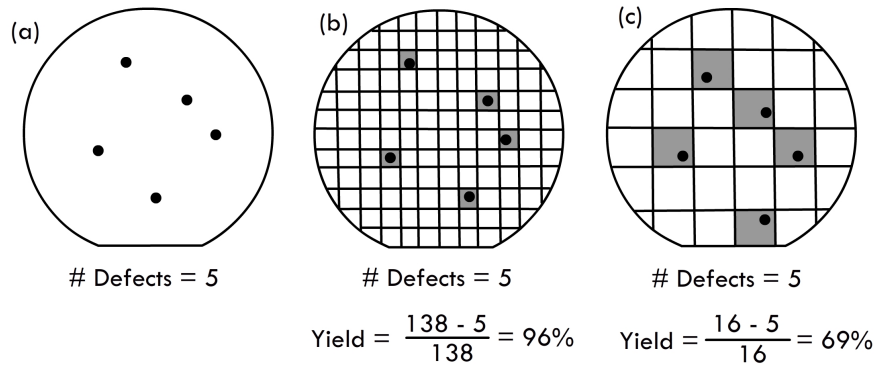


Figure 4: Effect of five random defects, i.e., background defect density on (a) wafer (b) wafer with smaller dies (c) wafers with larger dies. Larger die sizes will have lower yields. Adapted from *Microchip fabrication* - Peter van Zant.

7. Process cycle time - larger residence time for wafers in the fab can increase the defect density. Wafers that are stored in the fab between processes, can get contaminated and lower yield.

Die sizes and defect density are also related. For smaller die sizes, there are more dies per wafer. For the same number of defects, sort yield will be larger if the die size is smaller.

Packaging yield is related to the various steps in the packaging process. After sort, the individual wafers are separated and the packaged. After packaging the dies are again measured. The overall yield is then

$$\boxed{\text{Overall yield} = \text{wafer-fab yield} \times \text{sort yield} \times \text{Packaging yield}} \quad (3)$$

Thus, all three yield points determine the final output of the IC process.

$$p(k) = \frac{\exp(-m)m^k}{k!} \quad (4)$$

where m is n/N . Thus, the probability of a die having zero defects (good die) is then given by substituting $k = 0$ in equation 4.

$$p(0) = \exp(-m) = \text{Yield } (Y_r) \quad (5)$$

If we define chip defect density (D_0) to be the number of defects per die per unit area (i.e. $n/(N A)$) then

$$Y_r = \exp(-AD_0) \quad (6)$$

where A is the die area.